

$$\text{tr}(\lambda^{a_{\sigma_1}} \lambda^{a_{\sigma_2}} \dots \lambda^{a_{\sigma_{n-1}}} \lambda^{a_n})$$

Color Decomposition

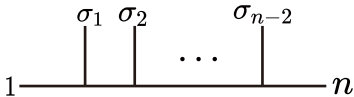


$$f^{dac} f^{cbe} - f^{dbc} f^{cae} = f^{abc} f^{dce}$$



$$f^{a_1 a_2 x_1} f^{x_1 a_3 x_2} \dots f^{x_{n-3} a_{n-1} a_n}$$

$$= (-i)^{n-2} \text{tr}(\lambda^{a_1} [\lambda^{a_2}, [\lambda^{a_3}, \dots, [\lambda^{a_{n-1}}, \lambda^{a_n}] \dots]])$$



$\sim$

$$f^{a_1 a_{\sigma_1} b_1} f^{b_1 a_{\sigma_2} b_2} \dots f^{b_{n-3} a_{\sigma_{n-2}} a_n}$$

DDM Basis